

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26 June 2026
Team ID	
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

AUTOMATED NETWORK REQUEST MANAGEMENT SYSTEM

DATA FLOW DIAGRAM (DFD)

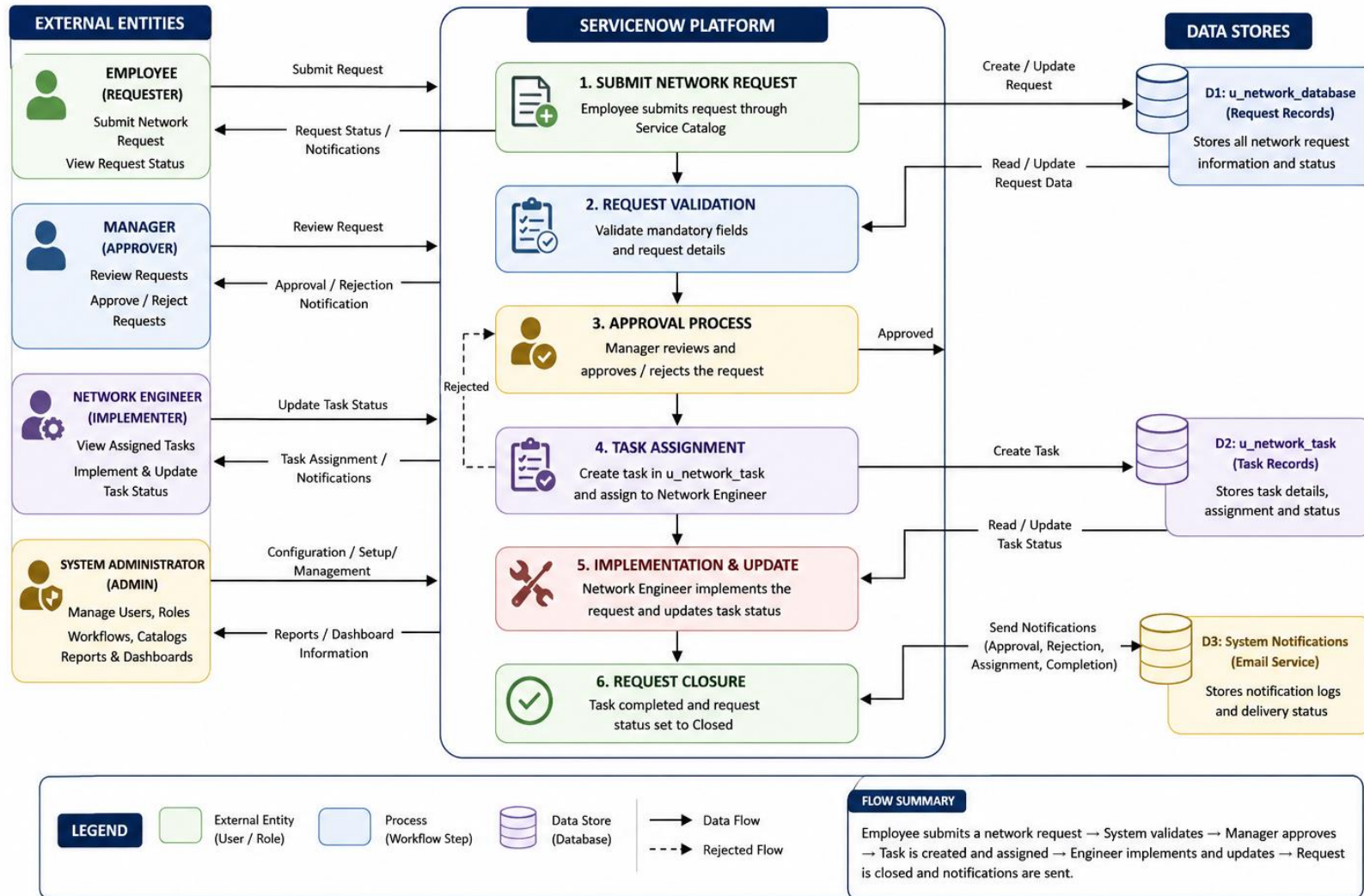


Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employees, Managers, and Network Engineers interact with the application through the Service Portal and request forms.	ServiceNow Service Portal, Catalog Item, Forms, Lists
2	Application Logic	Processes network request validation, request lifecycle management, and business logic execution.	Business Rules, Script Includes
3	Workflow Automation	Automates request approvals, task creation, status updates, and notifications.	Flow Designer
4	Database	Stores user information, network request records, implementation tasks, and request history.	sys_user, u_network_database, u_network_task
5	Notification Service	Sends automated notifications for request submission, approval, rejection, task assignment, and request completion.	ServiceNow Notifications, Email Templates
6	Security & Access Control	Provides secure access using role-based permissions for employees, managers, engineers, and administrators.	ACLs, Roles, User Criteria
7	Cloud Infrastructure	Hosts and manages the application on the ServiceNow cloud platform with centralized data management.	ServiceNow Cloud Platform

Table 2: Application Characteristics

Characteristic Description		Technology
Security	Provides secure authentication and role-based authorization for employees, managers, network engineers, and administrators.	ACLs, Roles, User Criteria
Scalability	Supports increasing numbers of network requests, users, and implementation tasks without significant performance degradation.	ServiceNow Platform

Characteristic Description		Technology
Availability	Ensures continuous access to the application whenever the ServiceNow instance is operational.	ServiceNow Cloud
Performance	Delivers efficient request processing through automated workflows, optimized forms, and background execution.	Flow Designer, Business Rules
Reliability	Maintains accurate request records, task information, approvals, and notifications while ensuring consistent workflow execution.	ServiceNow Database, Flow Designer